

PROJECT-NIRA: MICRO-PLASTIC DETECTION

1. Executive Summary

Project-Nira is an Open Source Hardware (OSHW) initiative designed to provide a cost-effective, real-time solution for detecting micro-plastics in water. This system addresses the critical need for decentralized environmental monitoring by offering a transparent alternative to proprietary sensing technologies.

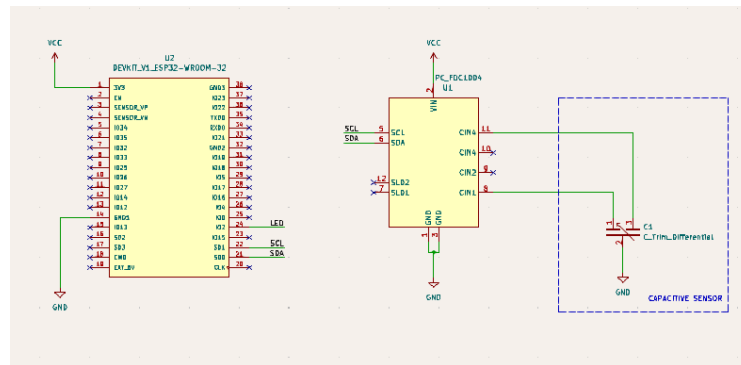


Fig 1.1 Circuit Diagram

The technical architecture leverages an ESP32 microcontroller and the ProtoCentral FDC1004 capacitance-to-digital converter, with data persistence managed through InfluxDB. By integrating these tools with 3D visualization protocols, Project-Nira demonstrates how Free and Open-Source Software (FOSS) principles can be applied to complex environmental challenges, ensuring public health data remains a shared resource rather than a private commodity.

2. Introduction: The Micro-plastic Crisis and Economic Barriers

Micro-plastic pollution has become a pervasive threat to global water security, yet monitoring efforts are frequently hampered by the "neoliberalism and privatization" of data.



Fig 2.1 Micro Plastic in water body

Current proprietary detection methods are often characterized by high costs and restrictive licensing, creating significant economic barriers for community-led environmental groups. Project-Nira counters this by adopting a FOSS/OSHW model, which prioritizes personal control, customizability, and freedom. This approach allows users to modify the system for specific local water conditions without

vendor interference. Furthermore, it addresses the security risks inherent in closed-source systems, where proprietary vendors may be pressured to include "backdoors" or covert features. In an OSHW model, the community can inspect and verify every line of code and hardware schematic.

Detection Methodology Comparison

Feature, Current Proprietary Methods, Project-Nira OSHW Approach

Cost, High acquisition/maintenance fees, "Low-cost, off-the-shelf components"

Licensing, "Restrictive, closed-source", Open-source (GPL) licensing

Customizability, Limited by vendor locks, Highly customizable by the end-user

Verification, "Opaque, ""black box"" testing", Community-verified; peer-reviewed

Security & Trust, "Risk of undisclosed ""backdoors""", Auditable source code; no hidden features

3. System Architecture & Design

The architecture of Project-Nira is built upon the synthesis of high-precision sensing hardware and a robust, open-standard repository structure.

3.1 Hardware Components

- **ESP32 Microcontroller:** Acts as the central processing unit, managing high-speed data acquisition from the sensors and facilitating wireless data transmission to the cloud.
- **ProtoCentral FDC1004:** This 4-channel capacitance-to-digital converter serves as the interface for the sensing electrodes, providing the high resolution required to detect minute changes in water permittivity.

3.2 Theory of Differential Capacitance Sensing

The system identifies micro-plastics based on the scientific principle of differential capacitance.

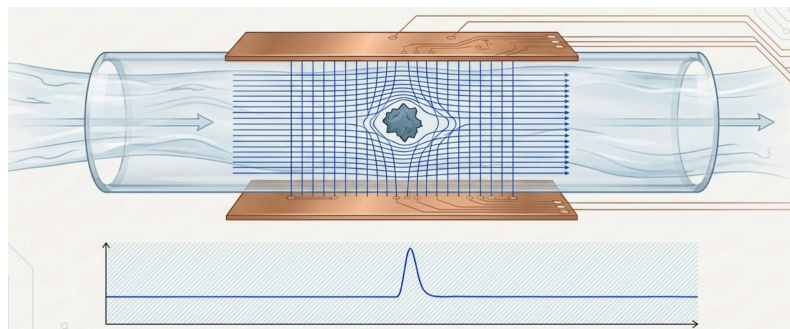


Fig 3.1 illustration of Differential Capacitance Sensing

Because micro-plastics possess a different relative permittivity compared to water, their presence within a sample alters the dielectric constant between the sensing electrodes. The FDC1004 detects these minute shifts in capacitance, allowing the system to categorize the concentration of contaminants in real-time.

3.3 Project Organisation & Standards

To maintain technical standards and encourage community contribution, the Project-Nira repository follows strict naming conventions. Per standard OSHW practices, all top-level files and folders use short lowercase names, with the exception of mandatory documentation files.

- `/src`: The core source files for the firmware and hardware logic.

- /docs: Reference data, hardware schematics, and technical manuals.
- LICENSE: The full text of the open-source license.
- README.md: The primary project introduction and setup guide.

4. Implementation & Data Pipeline

The data pipeline is designed to move seamlessly from raw physical sensing to high-level digital visualization.

4.1 Cloud Integration & Monitoring

The system employs a "Development to Deployment" pipeline, ensuring that the infrastructure remains scalable and reliable. Data is uploaded to **InfluxDB** for time-series analysis, supported by the following FOSS toolchain:

- **Version Control:** Git.
- **Containerization:** Docker & Docker Compose for environment consistency.
- **Orchestration:** Kubernetes (K3s) for managing decentralized deployments.
- **Monitoring:** Prometheus and Loki to track system health and log sensor events.

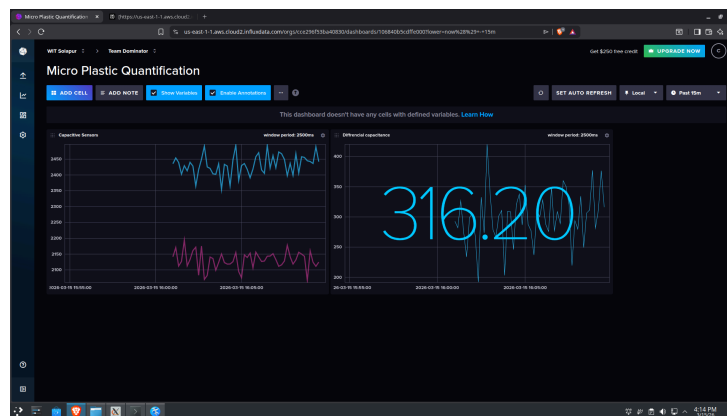


Fig 4.1 Dashboard

5. Open Source Philosophy & Governance

Project-Nira is an ideological extension of the digital health integration seen in initiatives like "Project Niramay," but applied to environmental health. It upholds the "Four Essential Freedoms":

- **Freedom 0:** Run the program for any purpose.
- **Freedom 1:** Study and change the code (access to source code is a precondition).
- **Freedom 2:** Redistribute copies.
- **Freedom 3:** Distribute modified versions to the community. Internal development follows an **InnerSource** approach, utilizing open collaboration and merit-based decisions. This model fosters "Faster Time-to-Market" and enhances knowledge management through community learning. To protect these freedoms, the project uses the **MIT License**, which permits broad commercial use while requiring only copyright preservation in distributions.

6. Conclusion: The Future of Decentralised Monitoring

Project-Nira represents a fundamental shift toward a decentralized environmental monitoring network. By providing the tools for micro-plastic detection as open assets, the project counters the monopolization of environmental data and fosters an "economy of abundance" for digital

knowledge. The success of such initiatives relies on broader institutional support. Following the **European Union's Open Source Strategy 2020-2023**, it is essential for governments to establish **Open Source Programme Offices** to manage and promote FOSS/OSHW. Supranational adoption of these standards—similar to the European Commission's move to release software under the EUPL—is the only way to ensure that the data protecting our public health remains transparent, auditable, and accessible to all.